Lesson 3: Take Your Shot

Godot Engine

VIDEO GAME CREATION

This week: spawning objects at runtime. Your job is to make your player shoot a projectile that flies up the screen and cleans itself up when it leaves - the first piece of your capstone's actual shooting mechanic.

Open your own [YourName]Game project from Lesson 2 - don't start a new one. Everything this week builds on the player object you already have.

Tip: this lesson's Assets folder has a ready-to-use placeholder projectile image (bullet_yellow.png) and a new sound effect (sfx_shoot.wav) - you don't need to find or make your own art today.

Setup

1. Open your [YourName]Game project from Lesson 2.
2. Create a new scene: a Sprite2D root node, rename it Bullet, and assign bullet_yellow.png (from this lesson's Assets folder) as its Texture.
3. Save this new scene as bullet.tscn in the project's root folder.
4. Attach a new script to the Bullet root, extends Sprite2D.

Level 1 - Minimal (MVP)

Pressing the fire key spawns one projectile that flies upward and disappears when it leaves the screen. This is a complete finish for today - reaching this is enough to move on to next week.

5. In the Bullet's own script, add an exported speed and a _process(delta) that moves it upward and frees itself once it's off-screen (code below).
6. Back in the Player's script (from Lesson 2), preload the bullet scene at the top and add an Input.is_action_just_pressed("ui_select") check in _process to instantiate it, place it at global_position (not position - global_position is the safe choice for matching an object's actual on-screen location no matter where it sits in the scene tree, and you'll keep using it for Enemy and Explosion in later lessons too), and add it as a child of get_parent() rather than the Player itself, so the bullet doesn't end up riding along attached to the Player - since Player is a child of Main (from Lesson 2's Setup), get_parent() resolves to Main, a real node you can see in the Scene tree, so the bullet lands as Player's sibling, not its passenger.
7. Press F5 (Run Project) rather than F6 this time - the bullet spawns via get_parent(), which only resolves correctly when Player is actually running as Main's child, not as its

own temporary scene root. Press Space (Godot's default ui_select action) to test - a bullet should spawn at the player and fly upward, then disappear near the top.

```
# bullet.tscn's script (attached to the Sprite2D root):
extends Sprite2D
var speed := 400.0

func _process(delta):
    position.y -= speed * delta
    if position.y < -20:
        queue_free()

# Player's script - add near the top and inside _process:
var bullet_scene := preload("res://bullet.tscn")

func _process(delta):
    if Input.is_action_just_pressed("ui_select"):
        var b := bullet_scene.instantiate()
        b.global_position = global_position
        get_parent().add_child(b)
```

Level 2 - Complete (Alpha/Beta)

Firing repeatedly produces multiple independent projectiles on screen at once, each moving and disappearing correctly, and rapid-fire is capped so holding the key down doesn't flood the screen.

- Test firing several times quickly - each press should produce its own independent bullet instance, all moving and freeing themselves correctly. Since each instantiate() call creates a fully separate node, this already works with no extra code.
- Add a cooldown so holding Space doesn't fire every frame: is_action_just_pressed() already only fires once per physical press (not once per frame held), so Level 1 is already immune to this specific bug - add a var fire_cooldown := 0.0 timer anyway if you want a slower, tunable fire rate (count it down by delta each frame, and only allow firing when it's <= 0, resetting it to e.g. 0.2 on each shot).

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Sound: add an AudioStreamPlayer2D node to the Player, set its Stream to sfx_shoot.wav, and call $AudioStreamPlayer2D.play() right next to the instantiate() call.
- Visual: give the Bullet its own modulate color or a small scale pulse that fades back to normal after a fraction of a second, so it reads clearly as "a shot" rather than a copy of the Player - not a permanent recolor.
- Structure: keep everything about how a bullet behaves - its own movement, its own off-screen cleanup - inside bullet.tscn's own script, and keep the Player's script limited to just spawning it (instantiate + position + cooldown + sound). Don't reach into a bullet's internals from the Player's script, or vice versa.

If Something's Wrong

- Nothing happens on Space: confirm Space is actually bound to the ui_select action in Project Settings > Input Map (it's a Godot default, but verify).

- Bullet spawns but doesn't move or never disappears: confirm the script is attached to bullet.tscn's own Sprite2D root, not left unattached, and that _process is spelled exactly right (case-sensitive).
- Error about instantiate() on a null value: the preload() path is wrong - confirm bullet.tscn was actually saved at res://bullet.tscn (or update the path to match where you saved it).
- Bullet spawns way off from the player - low and to the right, or some other consistent offset: confirm both sides of the assignment use global_position (b.global_position = global_position), not plain position - it's the safe choice for matching an object's actual on-screen location no matter where it sits in the scene tree. If bullets still drift or rotate along with the Player after that, confirm Player is actually a child of Main (from Lesson 2's Setup) - get_parent().add_child(b) only lands the bullet next to Player, as its sibling, if Main is really Player's parent. Also double check the bullet was added with get_parent().add_child(b), not get_tree().current_scene.add_child(b) - since Main is current_scene here, that line would still work, but it bypasses the actual parent-child relationship the rest of this lesson relies on.

Want More?

Student Resource List > Godot section, or docs.godotengine.org